:

Question 1

(a) Convert the following decimal value to binary and truncate at 4 bits after the radix. (9 Points)

$$58.3572_{10}$$

(b) Convert your answer from part (a) to hexadecimal (base-16). (5 Points)

(c) Perform the following multiplication in binary (base-2) and convert the result back into decimal. Recall: convert each operand to base-2 and reformulate the expression before performing the operation. (10 Points)

$$18_{10} \times 13_{10}$$

(d) If you are asked to perform $-150_{10} \times 4_{10}$ using only 2's complement representation; at least how many bits do the representations need to use to not encounter any overflow error? Hint: try to think of the result and how many bits are needed to represent that value. (6 Points)

Question 2

(a) Simplify each of the following expressions in one step by identifying and using a single theorem/law of Boolean algebra. Recall: expressions inside parentheses can be treated as

:

whole variables listed in your laws/theorems. (15 Points) i) $CF' + D(A + B') + (CF')'$

ii) $A(C + D)'E' + A(C + D)E'$

iii) $(B'CD + A)(B'CD + EF')$

(b) Prove the following equation. Hint: use the consensus theorem. (15 Points)

$$A'BC + A'C'D + AD = C'D + A'BC + AD + BD$$

Question 3

Q: Consensus theorem, validity of equations/expressions/statements Use

the following function for all parts of this question:

$F(W, X, Y, Z) = (XZ' \oplus Y)'(X'YZ + WY)$ (a) Draw a circuit for

the expression as it is given, without changing/simplifying. (8 Points)

(b) Eliminate the XOR and simplify to obtain the minimum SOP of the function, then draw a circuit implementation for the simplified function. Hint: simplify to 2 terms. (12 Points)

$(XZ' \oplus Y)'(X'YZ + WY)$

:

(c) Is the following statement always true? Explain your reasoning. (5 Points)

" If $AB = D$, then $ABC' = C'D$ "

Question 4

Consider the following situation for all parts of this question:
In an organization meeting, a proposal can be adopted if it: receives votes from a majority of its members or all of its officers, does not violate their rules, and does not exceed the budget.

(a) Identify all input variables and convert the statement into a Boolean function. (8 Points)

In an organization meeting, a proposal can be adopted if it: receives votes from a majority of its members or all of its officers, does not violate their rules, and does not exceed the budget.

F A

(b) Find the minterm expansion of the function; give your final answer in SOP form. (8 Points)

(c) Find the maxterm expansion of the function. (9 Points)